Estimation methods for migration flows from migrant stock data

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1 Introduction

1.1 Missing data on migration flows

We are interested in migration flows between all the countries of the world on the largest period possible.
In order to proceed to quantitative studies on these migration flows, we would ideally have complete origin and destination matrices covering all the countries of the world for a given period.
Via the Organisation for Economic Co-operation and Development (OECD), we can have access to net inflows and net outflows of foreign populations by country of the OECD. See `https://www.oecd.org/els/mig/keystat.htm`.
Via the UN Refugee Agency (UNHCR), one can have access to data on population displacements covering refugees, asylum-seekers. Data is available for disaggregation by sex. Nevertheless, this is only a part of all the migration flows. Access via the link: `https://www.unhcr.org/refugee-statistics/download/`.
Via the Integrated Modelling of European Migration (IMEM) database, we can obtain posterior distributions of migration flows covering 31 countries of the European Union and European Free Trade Association (EFTA) countries and the rest of the world.
There is a lack of data for migration flows concerning South-South flows.
That is when we use estimations strategies.

1.2 Estimation strategies of migration flows

Since data is not fully available on migration flows and when it is provided by statistical institutes these are not row data, there is a reflection to have on which estimation strategy is better to use rather than another one. The literature has developed methods based on migrant stock tables data.

1.2.1 Estimation based on migrant stock tables

Datasets The United Nations, via the International Migration Flows database provided by the Population Division (`https://www.un.org/development/desa/pd/content/international-migrant-stock`), affords access to migrant stock tables. There is also the World Bank that provide data with periods of ten years (`https://databank.worldbank.org/source/global-bilateral-migration`).

The literature used this data for indirect estimations of migration flows see (Abel et al., 2019, see)[and](Abel, 2013)[for example]. The data is available disaggregated by sex. The type of data is in discrete time with periods of five years between censuses moments. By row, information is provided on nationality or on place of birth while by column is provided information on the place of residence.

Estimation Methods As we will see in the core of the report, there are three main estimation strategies based on migrant stock table data. It is based on a few main operations: computing deltas, adjusting tables (a little of very basic matrix calculus), converting counts into rates (compute proportions), and proceeding to an iterative proportional fitting procedure (IPF). It is also based on Poisson regressions. The most difficult part is related to the interpretation of the results and the nature of what we estimate. It is also related to the identification of underlying assumptions behind any estimation strategy deployed.
Econometricians computed deltas and made some assumptions on the nature of the results to infer emigration flows estimates (Beine et al., 2011) (Bertoli and Moraga, 2015). The literature added assumptions to interpret in the case of negative differences to also be able to estimate return flows such that the total estimation is decomposed into estimates of return and emigration flows (Beine and Parsons, 2015).
With Dennett (2016) proposes a new approach and suggests estimating migration flows using migration rates estimates and total flows estimates.
Since Abel (2013) along with (Abel and Sander, 2014), a new approach based on migrant stock tables is provided. Nevertheless, there is an assumption maximizing the number of stayers that constraints to offer a "lower bound estimates" (Azose and Raftery, 2019). By the way, it leads to arbitrage between return and emigration flows during the estimation procedure and therefore to a potential under/over-estimation of the two.
Based on the recent literature, a strategy is adopted to provide estimates of transit, emigration and return flows composing the estimation of migration flows based on migrant stock data with (Azose and Raftery, 2019).
In this report, we propose to rethink the estimation strategy based on a simple delta between two migrant stock tables. These two are adjusted and a reflection is needed for the interpretation of the positive and negative entries. Different cases are needed to be analyzed. Due to the nature of these migrant stock tables, it will never be possible to estimate total flows but only a part of it since we can observe net flows conditional on place of birth only.

1.2.2 Estimations based on social networks' API

There exists another type of literature on estimation strategies related to migration flows. Nevertheless, we will not use them in the report. We briefly mention them in the introduction in the following two paragraphs.

Datasets New datasets are available thanks to social networks and lead to new ways to estimate migration flows ((Zagheni and Weber, 2012), (Zagheni et al., 2014)).

Estimation Methods In order to obtain more information that is given by the collection of tweets available for each user they got for their database, they use machine learning tools to extract more information. In (Zagheni et al., 2014), when they try to disaggregate by sex the collected tweeter's profiles whose tweets were downloaded via Tweeter's API with data of date/place of publication for example, they restrict the profile's dataset for which they also had a picture and then used Face++ machine learning's tool to identify whether these profiles were associated with males or females. They also used these pictures to classify by age via Face++'s black box.

2 Notations and definitions

2.1 General mathematical tools

$I(.)$ is the indicator function that is equal to 1 if the condition inside the parentheses is satisfied and 0 if not;

2.2 Migrant stock/flow matrices

2.2.1 Setting of a system

Consider a system $A_1, ..., A_n$ that will be our system of countries for which we have data from bilateral migrant stock tables S_t. They correspond to the row and column indices associated. It is a system since each A_i corresponds to an entity interacting with other A_j of the same collection. The interactions are migration flows. In practice we only measure net migration flows and of course with a lot of bias. All along the report, we will nevertheless proceed to examples with a small number of countries that will be from the system A,B,C;

closed system The system $A_1, ..., A_n$ is closed meaning that no interactions exist with other entities than those in the collection $A_1, ..., A_n$; This can be considered if the set of countries for which we have data is almost complete.

open system The system $A_1, ..., A_n$ is open meaning that interactions may exist with other entities; This can be considered if the set of countries for which we have data is incomplete.

2.3 Introduction to migrant stock/flow matrices

Let us introduce a pair of bilateral migrant stock tables S_t and S_{t+1} by providing an example.

Example

$$\mathbf{S_t} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 15 & 4 & 4 \\ 2 & 15 & 5 \\ 5 & 2 & 15 \end{pmatrix} \end{array}; \mathbf{S_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 20 & 8 & 6 \\ 4 & 23 & 5 \\ 6 & 4 & 18 \end{pmatrix} \end{array}$$

Rows refer to place of birth while columns refer to place of residence. We have three countries A, B and C. Entries are referring to native-born populations on the diagonal and foreign-born populations off the diagonal.
Now, let us add row margins:

$$\mathbf{S_t} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 15 & 4 & 4 & 23 \\ 2 & 15 & 5 & 22 \\ 5 & 2 & 15 & 22 \end{pmatrix} \end{array}; \mathbf{S_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 10 & 8 & 2 & 20 \\ 4 & 23 & 5 & 32 \\ 6 & 4 & 18 & 28 \end{pmatrix} \end{array}$$

You observe that the total by rows increased in A, B and C. This is the same as saying that the total population born in each country net increased between t and t+1.
This will be a problem for us to solve. Before inferring from these migrant stock tables any information on the movements of individuals between t and t+1, it will be necessary to adjust for demographic dynamics, otherwise, it could be a great source of bias. For example, looking at the difference in the cell (A,C) between t and t+1, you would notice a net decreasing stock of people born in A and living in C. You could interpret this phenomenon as flows between C and B of people born in A (transit flows) or as flows from C to A (return flows) but you could as well interpret this as simply death events. This is even more true that these migrant stock tables were not demographically adjusted.

Sometimes, even after demographically adjusting the migrant stock tables, we will not necessarily have equal row margins due to measurement errors for instance. In that case, we will still adjust for obtaining equal row margins in order to be able to interpret differences in cells as movements in columns between t and t+1.

Furthermore, let us introduce a birth-place-specific flow matrix:

$$\mathbf{F_t^k} := \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} F^k_{t;A,A} & F^k_{t;A,B} & F^k_{t;A,C} \\ F^k_{t;B,A} & F^k_{t;B,B} & F^k_{t;B,C} \\ F^k_{t;C,A} & F^k_{t;C,B} & F^k_{t;C,C} \end{pmatrix} \end{array}$$

where $F^k_{t;i,j}$ correspond to a flow from i to j between t and t+1 of people born in k. The entries are unknown and these are precisely what we would like to estimate. One trick would be, as it is done by abel2019climate, to use migrant stock data to estimate these values. Precisely, the procedure is to use migrant stock data as margins of this birth-place-specific flow matrix. Let us fix k = A and observe what happens:

$$\mathbf{F_t^A} := \begin{array}{c} \\ A \\ B \\ C \\ Total \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} F^A_{t;A,A} & F^A_{t;A,B} & F^A_{t;A,C} & 15 \\ F^A_{t;B,A} & F^A_{t;B,B} & F^A_{t;B,C} & 4 \\ F^A_{t;C,A} & F^A_{t;C,B} & F^A_{t;C,C} & 4 \\ 10 & 8 & 2 & \end{pmatrix} \end{array}$$

Notice that this matrix is not well defined since estimating inside values respecting these margins would imply that total by rows and total by columns differ. Therefore (Abel et al., 2019) proceed to demographic adjustments and other methods to obtain equal row margins.

We will do some operations on these migrant stock and flow matrices.

2.3.1 Operations on migrant stock/flow tables

On bilateral migrant stock tables

Demographic adjustments We will proceed to demographic adjustment of bilateral migrant stock tables. We introduce notations and operations: S^d_t Demographically adjusted migrant stock table at time t; Demographic adjustment, $S^d_t := S_t$ - D and $S^d_{t+1} := S_{t+1}$ - B where D and B are respectively Death and Birth matrices matching rows and columns indexes of our bilateral migrant stock tables.

Equal row margins We already introduced previously "equal row margins" as a condition for doing estimations based on migrant stock tables data. It is set for a couple (S_t, S_{t+1}) and it is equivalent to setting an identical total in population by place of birth between t and t+1. Concerning the matrices that were demographically adjusted and for which we adjusted to obtain equal row margins, we will have the following notations: S^{adj}_t, S^{adj}_{t+1}. The operations will be of two types using a closed or an open approach. The terms closed/open refer to the type of system we consider. This is detailed in the section "Logic of adjusting tables".

Difference operation Method for estimating migration flows from migrant stock data. It is defined by computing $\Delta S_{t+1} := S_{t+1}$ - S_t.

Migration rates Method for estimating migration flows from migrant stock data. It is based on converting off-diagonal values of a bilateral migrant stock table into composition data defined as migration rates.

Total flows estimation The estimation of total flows is used in the migration rates estimation method. This leads to operations on a pair of migrant stock tables. It applies a difference operation. Then there are multiple ways to construct an "estimation" of total flows.

On bilateral migrant flow tables Let F^k_t be a birth-place specific bilateral migrant flow table covering movements inside the system of the population born in k for flows between t and t+1.

$F^k_{i,j,t}$ entries i,j of the bilateral migrant flow table covering the sub-population of people born in k, with the period covering (t,t+1); We will also use the notation $f_{j->i|k}$ in a more explicit way out of a context of a bilateral migrant flow table with the interpretation of flow from j to i for population born in k; $\hat{F}^{k,D}_t$ Estimated bilateral migrant flow table by place of birth k; D corresponds to the method used among "M" and "I" respectively for Migrant

minimization and Independence;

Contingency table methods Methods for estimating migration flows from migrant stock data. It defines origin and destination matrices with margin constraints where margin data comes from migrant stock data. The rows correspond to the distribution of destinations conditional on place of birth and place of origin. The place of birth is fixed for each corresponding contingency table while rows are indexed by the collection of places of origin defined by our system A_1, ...,A_n.

2.4 Type of flows

2.4.1 Decomposition in transit, emigration, and return flows

Following (Azose and Raftery, 2019) decomposition of the migration flow, we will often refer to return flows, emigration flows, and transit flows. Return flows will refer to flows from a place of different from the place of birth back to the place of birth. Emigration flows will refer to flows from the place of birth to other places of the system. Transit flows will refer to the residual flows aside from return and emigration flows i.e. flows between two places different from the place of birth.

2.4.2 Cross flows by place of birth

We will often mention crossflows. Cross flows for a pair (A,B) of countries is simply the sum of flows from A to B and from B to A.
If we disaggregate by place of birth, we will notice that we do the sum of the emigration flows and the return flows when considering the sub-populations born in one of two countries considered i.e. A or B while we sum transit flows for all the other sub-populations considered.

3 Migrant-stock-tables

3.1 Migration - accounting definitions

We have the choice among three quantitative definitions of migration as an aggregate phenomenon.

There are three different questions we can ask to a random individual in a given country:
1) "Were you born abroad ?" (place of birth criterion; if yes, +1 foreign-born migrant; if no, +1 native-born population);
2) "Do you possess the nationality of the place in which you live?" (nationality criterion; if yes, +1 foreign population; if no, +1 national population);
3) "Did you recently move of place of residence for a duration of at least one year?" (change of place of residence criterion; if yes, +1 migrant population; if no, +1 non-migrant population).

3.2 Pros/Cons by criterion

Criterion	Pros	Cons
Place of birth	Comparability between countries; Availability;	Confound Old/Recent migrants;
Nationality	Availability	Lack of comparability (adjust for naturalization) and Confound Old/Recent migrants;
Change of place of residence by enquiries	Target recent migrants	Unavailability
Change of place of residence by new data sources	Estimate migration flows on a shorter scope	Methodologies still in development for data-processing ((Zagheni and Weber, 2012), (Zagheni et al., 2014))

Remarks The place of birth criterion is mostly used by the literature for proceeding to migration estimates. It is done by (Abel et al., 2019), (Parsons et al., 2007) for example.

3.3 Definition of a Bilateral migrant stock table

The literature calls the matrices we will look at "bilateral migrant stock tables".
- By bilateral it means that it is related to pair of countries data.
- By migrant it is in our case related to the place of birth criterion.
- By stock it is due to the time-dependency of the information we measure.
- By table it is related to its matrix form with rows indexed by our system of countries and columns indexed by the same system of countries. Rows are defined as places of birth and columns are defined as places of residence.

A couple of bilateral migrant stock table (S_t,S_{t+1}) in our report, will look like this:

$$\mathbf{S_t} = \begin{array}{c} \\ A \\ B \\ C \\ TR_t \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & TB_t \end{array} \\ \begin{pmatrix} S_t^{A,A} & S_t^{A,B} & S_t^{A,C} & TB_t^A \\ S_t^{B,A} & S_t^{B,B} & S_t^{B,C} & TB_t^B \\ S_t^{C,A} & S_t^{C,B} & S_t^{C,C} & TB_t^C \\ TR_t^A & TR_t^B & TR_t^C & N_t \end{pmatrix} \end{array} ; \mathbf{S_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \\ TR_{t+1} \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & TB_{t+1} \end{array} \\ \begin{pmatrix} S_{t+1}^{A,A} & S_{t+1}^{A,B} & S_{t+1}^{A,C} & TB_{t+1}^A \\ S_{t+1}^{B,A} & S_{t+1}^{B,B} & S_{t+1}^{B,C} & TB_{t+1}^B \\ S_{t+1}^{C,A} & S_{t+1}^{C,B} & S_{t+1}^{C,C} & TB_{t+1}^C \\ TR_{t+1}^A & TR_{t+1}^B & TR_{t+1}^C & N_{t+1} \end{pmatrix} \end{array}$$

It gives some information:
Summing over the rows: The total population by place of birth (TB_t^k);
Summing over the columns: The total population by place of residence (TR_{t+1}^k);
The values inside are the joint distribution of individuals by place of birth and place of residence at a given time;
Off-diagonal values: People who were born in another country than the one in which they live at time t. They are considered migrants.
Nevertheless, there are limits as far as off-diagonal values are concerned when it comes to their interpretation as

migrants:
The foreign-born population could be composed of a recent immigration population and an old immigration population. By recent immigration population, we think of people who emigrated recently from their place of birth or transited from another place of residence different than their place of birth to live in the considered place of residence.
By old immigration population (people who emigrated a long time ago from their place of birth to another place of residence in which they stayed before transiting again to the place of residence of interest for example) that might have, in between, acquired nationality and may not be considered - if by nationality criterion for example- as a migrant. There are of course numerous other examples.
In general, the problem is related to the lack of information on the previous duration of their stay at the given place of residence (short-term/long-term migrant?).

3.4 The setting to illustrate the different methods

$$\mathbf{S_t} = \begin{array}{c} \\ A \\ B \\ C \\ TR_t \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & TB_t \end{array} \\ \begin{pmatrix} 15 & 4 & 4 & 23 \\ 2 & 15 & 5 & 22 \\ 5 & 2 & 15 & 22 \\ 22 & 21 & 24 & 67 \end{pmatrix} \end{array} ; \mathbf{S_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \\ TR_{t+1} \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & TB_{t+1} \end{array} \\ \begin{pmatrix} 20 & 8 & 6 & 34 \\ 4 & 23 & 5 & 32 \\ 6 & 4 & 18 & 28 \\ 30 & 35 & 29 & 94 \end{pmatrix} \end{array}$$

Notice that, N_t and N_{t+1} can diverge due to demographic dynamics and flows with the entities outside of the system (open system) or due to measurement errors (closed system).
Notice that, populations by place of birth can also diverge from t to t+1 with here TB_t^k ¡ TB_{t+1}^k for all $k \in$ A,B,C because people migrated in/out of the system and because of newborns and death events.
The same by place of residence, due to demographic dynamics and movements in columns and with the rest of the world if open system.
For example, the number of individuals that are born in B residing in A at time t is 2 while 4 at time t+1.
(D_t^A, D_t^B, D_t^C) = (5,4,3);
This corresponds to the vector of total death by place of residence between t and t+1.
(b_A, b_B, b_C) := (13, 10, 10);
This corresponds to the vector of birth by place of residence between t and t+1.
Hence the next section covering adjustments of bilateral migrant stock tables. It covers demographic adjustments as well as row adjustments.

4 Logic of adjusting tables before estimation

4.1 Motivation

About the literature Adjusting tables before estimation is done by the literature (Abel et al., 2019), (Abel, 2013) for contingency table methods. According to the formal presentation of difference methods in his paper of 2019 and the R code + data shared on Figshare, Abel did not adjust the migrant stock tables before using difference methods or migration rate methods. Nevertheless, for reverse negative, the starting literature (Beine and Parsons, 2015) did not indeed proceed to migrant stock table demographic adjustment. The following methods are from the literature: (Abel, 2013) for demographic adjustments and open approach, (Abel and Sander, 2014) for closed approach.

In a general approach Adjusting migrant stock tables seems relevant whatever the method used for estimating migrant stock tables.

Since the goal is to estimate migration flows from differences in stocks of foreign-born populations between two-time units, it would be a good option to systematically adjust for demographic dynamics, not only for contingency table methods.

4.2 Demographic adjustment methods

We will illustrate the demographic adjustment methods on the first bilateral migrant stock table defined in the previous section.

4.2.1 Logic of demographic adjustment methods

Between t and t+1, birth and death events occurred which impacted the entries of the stock table at time t+1. Birth events correspond to newborns between t and t+1 and death events correspond to the total number of dead people by place of birth and place of residence between t and t+1 of individuals living at t. Then logically the adjustment is done such that we remove at time t+1 from S_{t+1} birth data as an added quantity between t and t+1 while we remove at time t from S_t death data as a removed quantity between t and t+1 to neutralize these dynamics in explaining the difference between S_t and S_{t+1}.

We proceed to demographic adjustments by removing at S_t the total death while removing at S_{t+1} total birth. Concerning birth adjustment, the idea is just that between t and t+1 people popped up that were not here and modified the total population between t and t+1 (births) therefore we removed them when we count them i.e. at time t+1. Concerning death adjustment, the idea is just that between t and t+1 people died between t and t+1 therefore were not counted at time t+1 but were counted at time t. We remove them directly at time t. In a sense, the adjustment is following the idea that we do not count people at time t that will not be counted at time t+1 and we do not count people at time t+1 that were not counted at time t. The goal is to explain differences in columns by migration flows and not have differences due to people counted but were not there at the time t or diminishing stocks due to death events.

4.2.2 Birth data adjustment

Birth data adjustment is done by withdrawing to the migrant stock table at time t+1 the newborns. Since available data is only on total birth by place of residence, we assume that newborns did not move in this interval of time - hence withdrawing them with a diagonal matrix.

$$\mathbf{S^d_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \end{array}\!\!\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 20 & 8 & 6 \\ 4 & 23 & 5 \\ 6 & 4 & 18 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array}\!\!\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} b_A & 0 & 0 \\ 0 & b_B & 0 \\ 0 & 0 & b_C \end{pmatrix} \end{array}$$

Recall that $(b_A, b_B, b_C) := (13, 10, 10)$, you obtain:

$$\mathbf{S^d_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \\ Total \end{array}\!\!\begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 7 & 8 & 6 & 21 \\ 4 & 13 & 5 & 22 \\ 6 & 4 & 8 & 18 \\ 17 & 25 & 19 & 61 \end{pmatrix} \end{array}$$

We will use the information on $(TB_t^A, TB_t^B, TB_t^C) = (21,22,28)$ later with the next step in the adjustment of the stock tables to obtain equal row margins.

4.2.3 Death data adjustment

Death data adjustment is done by withdrawing to the migrant stock table at time t the number of dead individuals between t and t+1. Since available data is only on total death by place of residence, we remark that $D_t^k = \sum_i D_t^{i,k}$, we can estimate by a proportion method the values $D_t^{i,k}$ for all i $\in$ A,B,C. This is done as follows:

$$\mathbf{S_t^d} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 15 & 4 & 4 \\ 2 & 15 & 5 \\ 5 & 2 & 15 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} \hat{D}_t^{A,A} & \hat{D}_t^{A,B} & \hat{D}_t^{A,C} \\ \hat{D}_t^{B,A} & \hat{D}_t^{B,B} & \hat{D}_t^{B,C} \\ \hat{D}_t^{C,A} & \hat{D}_t^{C,B} & \hat{D}_t^{C,C} \end{pmatrix} \end{array}$$

where $\hat{D}$ is obtained by proportion method i.e. $\hat{D}_t^{i,j} = D_t^j \times \frac{S_t^{i,j}}{\sum_i S_t^{i,j}}$ $\forall (i,j) \in (A,B,C)^2$.

Recall that (D_t^A, D_t^B, D_t^C) = (5,4,3).

Then we have:

$$\hat{\mathbf{D}} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 5 \times \frac{15}{22} & 4 \times \frac{4}{21} & 3 \times \frac{4}{24} \\ 5 \times \frac{2}{22} & 4 \times \frac{15}{21} & 3 \times \frac{5}{24} \\ 5 \times \frac{5}{22} & 4 \times \frac{2}{21} & 3 \times \frac{15}{24} \end{pmatrix} \end{array} \Rightarrow \mathbf{S_t^d} = \begin{array}{c} \\ A \\ B \\ C \\ Total \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 & 18.2 \\ 4.5 & 12.2 & 4.375 & 21.075 \\ 4 & 1.6 & 13.125 & 18.725 \\ 20 & 17 & 21 & 58 \end{pmatrix} \end{array}$$

In this part, we have briefly presented the logic behind demographic adjustments of bilateral migrant stock tables as well as illustrating the methods used with data real constraints (available birth/death data by place of residence only).

4.3 Final adjustment for equal row margins

Recall that the goal of equal row margins is to have the same total population by place of birth between t and t+1.

4.3.1 Difference in row margins

$$\mathbf{TB_{t+1}} - \mathbf{TB_t} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} Total \\ \begin{pmatrix} S_{t+1,A.}^d \\ S_{t+1,B.}^d \\ S_{t+1,C.}^d \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} Total \\ \begin{pmatrix} S_{t,A.}^d \\ S_{t,B.}^d \\ S_{t,C.}^d \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} Delta \\ \begin{pmatrix} \Delta S_{t+1,A.}^d \\ \Delta S_{t+1,B.}^d \\ \Delta S_{t+1,C.}^d \end{pmatrix} \end{array}$$

Sticking to our previous example,

$$\mathbf{TB_{t+1}} - \mathbf{TB_t} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} Total \\ \begin{pmatrix} 21 \\ 22 \\ 18 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} Total \\ \begin{pmatrix} 18.2 \\ 21.075 \\ 18.725 \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} Delta \\ \begin{pmatrix} 2.8 \\ 0.925 \\ -0.725 \end{pmatrix} \end{array}$$

where TB_{t+1}, TB_t are here the restrictions of the demographically adjusted migrant stock tables to their row margins.

4.3.2 Simple scaling method

If no assumption is made on explaining the residual differences in total populations by place of birth between t and t+1, the literature simply rescales the two demographically adjusted stock tables to obtain equal row margins. It requires only to apply a rule of three hence we skip details and provide directly the resulting adjusted stock tables with our previous example.

They rescale this way:

$$\mathbf{S_t^{adj}} := \begin{pmatrix} \alpha_A^t & 0 & 0 \\ 0 & \alpha_B^t & 0 \\ 0 & 0 & \alpha_C^t \end{pmatrix} . \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} S^d_{t;A,A} & S^d_{t;A,B} & S^d_{t;A,C} \\ S^d_{t;B,A} & S^d_{t;B,B} & S^d_{t;B,C} \\ S^d_{t;C,A} & S^d_{t;C,B} & S^d_{t;C,C} \end{pmatrix} \end{array}$$

such that, by summing over the row k, we obtain the associated value: $\frac{(\sum_i S^d_{t;k,i} + \sum_i S^d_{t;k,i})}{2}$.

The same thing is applied for S^d_{t+1} consistently with the objective to obtain equal row margins.

Indeed,

$$\mathbf{S_{t+1}^{adj}} := \begin{pmatrix} \alpha_A^{t+1} & 0 & 0 \\ 0 & \alpha_B^{t+1} & 0 \\ 0 & 0 & \alpha_C^{t+1} \end{pmatrix} . \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} S^d_{t+1;A,A} & S^d_{t+1;A,B} & S^d_{t+1;A,C} \\ S^d_{t+1;B,A} & S^d_{t+1;B,B} & S^d_{t+1;B,C} \\ S^d_{t+1;C,A} & S^d_{t+1;C,B} & S^d_{t+1;C,C} \end{pmatrix} \end{array}$$

such that, by summing over the row k, we obtain the associated value: $\frac{(\sum_i S^d_{t+1;k,i} + \sum_i S^d_{t+1;k,i})}{2}$.

We provide the example next page.

Applied scaling demographically adjusted migrant stock tables:

$$\mathbf{S_t^{adj}} := \begin{pmatrix} \alpha_A^t & 0 & 0 \\ 0 & \alpha_B^t & 0 \\ 0 & 0 & \alpha_C^t \end{pmatrix} . \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 & 18.2 \\ 4.5 & 12.2 & 4.375 & 21.075 \\ 4 & 1.6 & 13.125 & 18.725 \end{pmatrix} \end{array}$$

$$\mathbf{S_{t+1}^{adj}} := \begin{pmatrix} \alpha_A^{t+1} & 0 & 0 \\ 0 & \alpha_B^{t+1} & 0 \\ 0 & 0 & \alpha_C^{t+1} \end{pmatrix} . \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 7 & 8 & 6 & 21 \\ 4 & 13 & 5 & 22 \\ 6 & 4 & 8 & 18 \end{pmatrix} \end{array}$$

Then, simply by computing α_i^h for i in (A,B,C) and h in (t, t+1) with $\alpha_i^h = \frac{1}{2} + \frac{TB_{t+1}^i}{2TB_t^i}$, we obtain:
$(\alpha_i^t)_{i\in(A,B,C)} = (1.07,\ 0.93,\ 1.046)$
$(\alpha_i^{t+1})_{i\in(A,B,C)} = (0.93,\ 0.89,\ 1.09)$

Which yields:

$$S_{t+1}^{adj} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 6.51 & 7.44 & 5.58 & 19.6 \\ 3.56 & 11.57 & 4.45 & 21.53 \\ 6.54 & 4.36 & 8.72 & 18.36 \end{pmatrix} \end{array} S_t^{adj} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 12.305 & 3.424 & 3.745 & 19.6 \\ 4.185 & 11.346 & 4.068 & 21.53 \\ 4.9 & 1.67 & 13.73 & 18.36 \end{pmatrix} \end{array}$$

Where we finally obtain equal row margins.

4.3.3 Closed approach: Diagonal adjustment + IPF

$$\Delta N_{t+1} = \mathbf{N_{t+1}} - \mathbf{N_t}$$

<u>Step 1</u>: Removing on the diagonal to obtain equal total populations between t and t+1

$$\mathbf{S_t^{(1)}} := \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} S^d_{t;A,A} & S^d_{t;A,B} & S^d_{t;A,C} \\ S^d_{t;B,A} & S^d_{t;B,B} & S^d_{t;B,C} \\ S^d_{t;C,A} & S^d_{t;C,B} & S^d_{t;C,C} \end{pmatrix} \end{array} + \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} \frac{\Delta N_{t+1}}{2} * \frac{S^d_{t;A,A}}{\sum_k S^d_{t;k,k}} & 0 & 0 \\ 0 & \frac{\Delta N_{t+1}}{2} * \frac{S^d_{t;B,B}}{\sum_k S^d_{t;k,k}} & 0 \\ 0 & 0 & \frac{\Delta N_{t+1}}{2} * \frac{S^d_{t;C,C}}{\sum_k S^d_{t;k,k}} \end{pmatrix} \end{array}$$

$$\mathbf{S_{t+1}^{(1)}} := \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} S^d_{t+1;A,A} & S^d_{t+1;A,B} & S^d_{t+1;A,C} \\ S^d_{t+1;B,A} & S^d_{t+1;B,B} & S^d_{t+1;B,C} \\ S^d_{t+1;C,A} & S^d_{t+1;C,B} & S^d_{t+1;C,C} \end{pmatrix} \end{array} - \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} \frac{\Delta N_{t+1}}{2} * \frac{S^d_{t+1;A,A}}{\sum_k S^d_{t+1;k,k}} & 0 & 0 \\ 0 & \frac{\Delta N_{t+1}}{2} * \frac{S^d_{t+1;B,B}}{\sum_k S^d_{t+1;k,k}} & 0 \\ 0 & 0 & \frac{\Delta N_{t+1}}{2} * \frac{S^d_{t+1;C,C}}{\sum_k S^d_{t+1;k,k}} \end{pmatrix} \end{array}$$

<u>Step 2</u>: IPFP on both $S_t^{(1)}$ and $S_{t+1}^{(1)}$ with maintained columns totals and modified row margins.

$$\mathbf{TB_{t+1}} - \mathbf{TB_t} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} residual \\ \begin{pmatrix} TB^A_{t+1} - TB^A_t \\ TB^B_{t+1} - TB^B_t \\ TB^C_{t+1} - TB^C_t \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} residual \\ \begin{pmatrix} \Delta TB^A_{t+1} \\ \Delta TB^B_{t+1} \\ \Delta TB^C_{t+1} \end{pmatrix} \end{array}$$

$$\begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 & 18.2 \\ 4.5 & 12.2 & 4.375 & 21.075 \\ 4 & 1.6 & 13.125 & 18.725 \end{pmatrix} \end{array}$$

$$\begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 7 & 8 & 6 & 21 \\ 4 & 13 & 5 & 22 \\ 6 & 4 & 8 & 18 \end{pmatrix} \end{array}$$

Then, simply by computing α_i^h for i in (A,B,C) and h in (t, t+1) with $\alpha_i^h = \frac{1}{2} + \frac{TB_{t+1}^i}{2TB_t^i}$, we obtain:
$(\alpha_i^t)_{i\in(A,B,C)} = (1.07,\ 0.93,\ 1.046)$
$(\alpha_i^{t+1})_{i\in(A,B,C)} = (0.93,\ 0.89,\ 1.09)$

Which yields:

$$S_{t+1}^{adj} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 6.51 & 7.44 & 5.58 & 19.6 \\ 3.56 & 11.57 & 4.45 & 21.53 \\ 6.54 & 4.36 & 8.72 & 18.36 \end{pmatrix} \end{array} S_t^{adj} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} 12.305 & 3.424 & 3.745 & 19.6 \\ 4.185 & 11.346 & 4.068 & 21.53 \\ 4.9 & 1.67 & 13.73 & 18.36 \end{pmatrix} \end{array}$$

4.3.4 Open approach: Adjusting for net flows with outside countries

Remember our residuals:

$$\mathbf{TB_{t+1}} - \mathbf{TB_t} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} residual \\ \begin{pmatrix} 2.8 \\ 0.925 \\ -0.725 \end{pmatrix} \end{array}$$

Then the strategy is to assume that, in an open system, strictly positive (negative) differences in total population by place of birth are the result of outflows from the system to other entities strictly less (greater) than return flows from these other entities to our system between t and t+1.

Adjustment for positive residuals: The positive residuals are interpreted as net return flows for all k in (A,B,C). We adjust for it by removing at S^d_{t+1} following the accounting identity logics at the corresponding row with a proportion method (Abel, 2013).

$$\mathbf{S^{adj}_{t+1}} = \begin{pmatrix} 7 & 8 & 6 \\ 4 & 13 & 5 \\ 6 & 4 & 8 \end{pmatrix} - \begin{pmatrix} 2.8 \times \frac{7}{21} & 2.8 \times \frac{8}{21} & 2.8 \times \frac{6}{21} \\ 0.925 \times \frac{4}{22} & 0.925 \times \frac{13}{22} & 0.925 \times \frac{5}{22} \\ 0 & 0 & 0 \end{pmatrix} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & TB_{t+1} \end{array} \\ \begin{pmatrix} 6.076 & 6.937 & 5.216 & 18.229 \\ 3.828 & 12.441 & 4.785 & 21.054 \\ 6 & 4 & 8 & 18 \end{pmatrix} \end{array}$$

Adjustment for negative residuals: The negative residuals are interpreted as net emigration flows (emigration in terms of In/Out entities) for all k in (A,B,C). We adjust for it by removing at S^d_t following the accounting identity logics at the corresponding row with a proportion method (Abel, 2013).

$$\mathbf{S^{adj}_{t}} = \begin{pmatrix} 11.5 & 3.2 & 3.5 \\ 4.5 & 12.2 & 4.375 \\ 4 & 1.6 & 13.125 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0.725 \times \frac{4}{18.75} & 0.725 \times \frac{1.6}{18.75} & 0.725 \times \frac{13.125}{18.75} \end{pmatrix}$$

$$\mathbf{S^{adj}_{t}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & TB_t \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 & 18.2 \\ 4.5 & 12.2 & 4.375 & 21.075 \\ 3.84 & 1.536 & 12.6 & 17.976 \end{pmatrix} \end{array}$$

4.4 Remarks

Adjusting as a source of bias in the case of sparse migrant stock tables

Uniform demographic adjustment In the case of uniform imputation when doing demographic adjustments, due to sparsity of matrices, it would automatically increase the difference between t and t+1 for off-diagonal values with entries zero at time t and t+1 and it would artificially count death adjustment's values for net emigration flows.

In the case of uniform imputation when doing demographic adjustments, adjusting for death data would mod-

ify the entries of S_t and put negative values since most of the entries are zeros. Then we would have negative values for foreign-born populations. It would then be impossible to extract from it any empirical distributions due to negative values.

Therefore, uniform imputation is not a good approach if we intend on adjusting demographically the bilateral migrant stock tables with a property of sparsity.

Proportional demographic adjustment In the case of proportional imputation when doing demographic adjustments - with the sparsity property, what can happen is the following:

- Zero entries at both t: No modification with adjustment for death data since proportion method;
- Zero entries only at t+1: Modification of the values at time t and since zero entries at t+1 yield negative difference that, in a closed system and equal row margins, could be interpreted as net return flows;
- Zero entries at t+1 and ϵ values at t: adjustment for death data leads to negative values on the adjusted cell which then creates, when doing the difference between the two adjusted stock tables with equal row margins, net emigration flows; It would modify the procedure of migration rates since we would have off-diagonal values with negative entries;
- Other cases might be relevant to emphasize.

5 Bilateral migrant flow tables: Decomposition into Emigration, Return and Transit flows

Let us define an origin and destination matrix with our system (A,B,C):

$$\mathbf{F_t} := \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} F_{A,A} & F_{A,B} & F_{A,C} \\ F_{B,A} & F_{B,B} & F_{B,C} \\ F_{C,A} & F_{C,B} & F_{C,C} \end{pmatrix} \end{array}$$

Each entry corresponds to flows between i and j with by rows place of origin and by column place of residence. The strategy to estimate these entries is to decomposed by place of birth as follows:
$F_t := \sum_k F_t^k$
where,

$$\mathbf{F_t^k} := \begin{array}{c} \\ A \\ B \\ C \\ \\ \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & & \end{array} \\ \begin{pmatrix} F_{A,A}^k & F_{A,B}^k & F_{A,C}^k & | & S_{k,A}^t \\ F_{B,A}^k & F_{B,B}^k & F_{B,C}^k & | & S_{k,B}^t \\ F_{C,A}^k & F_{C,B}^k & F_{C,C}^k & | & S_{k,C}^t \\ - & - & - & & \\ S_{k,A}^{t+1} & S_{k,B}^{t+1} & S_{k,C}^{t+1} & & \end{pmatrix} \end{array}$$

where $F_{i,j}^k$ is unknown.

Interpretation of $F_{i,j}^k$: Take for instance with k = A and define the following origin and destination matrix birth-place specific:

$$\mathbf{F_t^A} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} Z_{A->A|A} & E_{A->B|A} & E_{A->C|A} & S_{t;A,A}^{adj} \\ R_{B->A|A} & Z_{B->B|A} & T_{B->C|A} & S_{t;A,B}^{adj} \\ R_{C->A|A} & T_{C->B|A} & Z_{C->C|A} & S_{t;A,C}^{adj} \\ S_{t+1;A,A}^{adj} & S_{t+1;A,B}^{adj} & S_{t+1;A,C}^{adj} & \end{pmatrix} \end{array}$$

where values on the diagonal correspond to stayers and off the diagonal correspond to transit, emigration, and return flows in particular flows with letters "E" correspond to emigration flows; with letters "T" correspond to transit flows; with letters "R" correspond to return flows.

Interpretation: Therefore, $E_{A->B|A}$ corresponds to flows between A and B for people born in A (emigration flows) while $R_{C->A|A}$ corresponds to flows between C and A for people born in A (return flows). Another example, $Z_{A->A|A}$ corresponds to stayers and $T_{B->C|A}$ corresponds to transit flows from B to C for people born in A.

The matrix is constructed as follows:
Summing over the row i yields the total population born in k living in i at time t hence $S_t^{k,i}$.
Summing over column j yields the total population born in k living in j at time t+1 hence $S_{t+1}^{k,i}$.
Then, doing the same for F_t^B and F_t^C, you obtain:

$$\mathbf{F_t^B} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} Z_{A->A|B} & R_{A->B|B} & T_{A->C|B} & S^{adj}_{t;B,A} \\ E_{B->A|B} & Z_{B->B|B} & E_{B->C|B} & S^{adj}_{t;B,B} \\ T_{C->A|B} & R_{C->B|B} & Z_{C->C|B} & S^{adj}_{t;B,C} \\ S^{adj}_{t+1;B,A} & S^{adj}_{t+1;B,B} & S^{adj}_{t+1;B,C} & \end{pmatrix} \end{array}$$

$$\mathbf{F_t^C} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} Z_{A->A|C} & T_{A->B|C} & R_{A->C|C} & S^{adj}_{t;C,A} \\ T_{B->A|C} & Z_{B->B|C} & R_{B->C|C} & S^{adj}_{t;C,B} \\ E_{C->A|C} & E_{C->B|C} & Z_{C->C|C} & S^{adj}_{t;C,C} \\ S^{adj}_{t+1;C,A} & S^{adj}_{t+1;C,B} & S^{adj}_{t+1;C,C} & \end{pmatrix} \end{array}$$

Which implies that in fact entries of F_t is:

$$\mathbf{F_t} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} Z_{A->A|A} + Z_{A->A|B} + Z_{A->A|C} & E_{A->B|A} + R_{A->B|B} + T_{A->B|C} & E_{A->C|A} + R_{A->C|C} + T_{A->C|B} \\ E_{B->A|B} + R_{B->A|A} + T_{B->A|C} & Z_{B->B|B} + Z_{B->B|A} + Z_{B->B|C} & E_{B->C|B} + R_{B->C|C} + T_{B->C|A} \\ E_{C->A|C} + R_{C->A|A} + T_{C->A|B} & E_{C->B|C} + R_{C->B|B} + T_{C->B|A} & Z_{C->C|C} + Z_{C->C|B} + Z_{C->C|A} \end{pmatrix} \end{array}$$

where on the diagonal we have the stayers and the flows.
A key remark is that we use margins of migrant stock tables.
If we add the margins, we see the link with our migrant stock tables.

$$\begin{array}{c} \\ A \\ B \\ C \\ Total \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & Total \end{array} \\ \begin{pmatrix} Z_{A->A|A} + Z_{A->A|B} + Z_{A->A|C} & E_{A->B|A} + R_{A->B|B} + T_{A->B|C} & E_{A->C|A} + R_{A->C|C} + T_{A->C|B} & \sum_k S^{adj}_{t;k,C} \\ E_{B->A|B} + R_{B->A|A} + T_{B->A|C} & Z_{B->B|B} + Z_{B->B|A} + Z_{B->B|C} & E_{B->C|B} + R_{B->C|C} + T_{B->C|A} & \sum_k S^{adj}_{t;k,C} \\ E_{C->A|C} + R_{C->A|A} + T_{C->A|B} & E_{C->B|C} + R_{C->B|B} + T_{C->B|A} & Z_{C->C|C} + Z_{C->C|B} + Z_{C->C|A} & \sum_k S^{adj}_{t;k,C} \\ \sum_k S^{adj}_{t+1;k,A} & \sum_k S^{adj}_{t+1;k,B} & \sum_k S^{adj}_{t+1;k,C} & \end{pmatrix} \end{array}$$

6 Migrant-stock-table-based estimation methods

6.1 Difference Methods

Let us now present the difference methods for estimating migration flows out of adjusted bilateral migrant stock tables.

6.1.1 "Drop Negative"

Presentation and example The "drop negative" method ((Beine et al., 2011) ; (Bertoli and Moraga, 2015) ; (Abel et al., 2019)) is one way to deal with the difference between a pair of migrant stock tables.
We provide an example:

$$(\mathbf{\Delta S^d_{t+1}} - \mathbf{diag}(\mathbf{\Delta S^d_{t+1}}))_+ = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 8 & 6 \\ 4 & 0 & 5 \\ 6 & 4 & 0 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 3.2 & 3.5 \\ 4.5 & 0 & 4.375 \\ 4 & 1.6 & 0 \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 4.8 & 2.5 \\ -0.5 & 0 & 0.625 \\ 2 & 2.4 & 0 \end{pmatrix} \end{array}$$

Pros and Cons

Difference Method	Pros	Cons
"Drop negative"	Partial Emigration Flows Estimation	No return neither transit flows estimated + strong assumption for emigration flows

6.1.2 "Reverse Negative"

Presentation and example We simply, instead of dropping negative values, kept them and add them to the previous estimated matrix at its symmetric coordinate.

$$(\mathbf{\Delta S^d_{t+1}})_+ + (\mathbf{\Delta S^d_{t+1}})^{\mathbf{T}}_- - \mathbf{diag}((\mathbf{\Delta S^d_{t+1}}) = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 5.3 & 2.5 \\ 0 & 0 & 0.625 \\ 2 & 2.4 & 0 \end{pmatrix} \end{array}$$

Pros and Cons

Difference Method	Pros	Cons
"Reverse negative"	Partial Emigration flows + Return Flows Estimation	No transit flows estimated + strong assumption for emigration and return flows

6.1.3 Alternative Approach

The alternative is to focus on the meaning of positive and negative differences and finding the best interpretation possible of the net inflows (respectively net outflows) associated based on the information available in the best case of completely adjusted bilateral migrant stock tables (demographically + equal row margins). If we try to decompose what we really compute via differencing two stock tables, assuming we have demographically adjusted for death events and obtained equal row margins and assuming a closed system, we would have the following:

$$\Delta S^{adj}_{t+1} := (E^{i->j|i}_t + \textstyle\sum_{l\neq(i,j)}(T^{l->j|i}_t - T^{j->l|i}_t) - R^{j->i}_t)_{i\neq j}.$$

where $\Delta S^{adj}_{t+1} = S^{adj}_{t+1} - S^{adj}_t$, $E^{i->j|i}_t$ is an emigration flow from i to j i.e. conditional on the place of birth i between t and t+1, $T^{j->k|i}_t$ is a transit flow from j to k covering the population of individuals born in i, R^i_t is a return flow from j to i i.e. covering sub-population of individuals born in i going back to i between t and t+1.

Nevertheless, following the literature (Abel et al., 2019), (Beine and Parsons, 2015), the migrant stock tables are not demographically adjusted and the population by place of birth may be different between t and t+1 due to measurement errors or interactions with other entities than inside the system considered. This is explicitly acknowledged in (Beine and Parsons, 2015) emphasized and implicitly underlined in (Abel et al., 2019) by theoretically illustrating the methods with adjusted stock tables even though in practice these are not adjusted. This is yet to determine why they are not adjusted, what would be the pros and cons, and what would be the implications. But why not at least adjust the best as possible? Nevertheless, the procedure of demographic adjustments came with (Abel, 2013) i.e. later after the first papers were published on difference-methods.

Therefore, without adjusting, it would look like this:
$(\Delta S^{adj}_{t+1})_{i\neq j} := (E^{i->j|i}_t + \sum_{l\neq(i,j)}(T^{l->j|i}_t - T^{j->l|i}_t) - R^{j->i}_t - D^{i,j}_t) + \epsilon^{i,j}_t)_{i\neq j}$

where we added $-D_t^{i,j}$ accounting for demographic dynamics of the foreign-born population and $\epsilon_t^{i,j}$ accounting for residual differences in row margins between t and t+1 to be adjusted.

The literature first made an assumption about positive differences and did not count the negative differences and then made an assumption about the negative differences. Abel calls these methods respectively "drop negative" and "reverse negative" (Abel et al., 2019).

6.1.4 Positive differences: Net inflows

This is what we have in the case of positive differences on off-diagonal values:

$E_t^{i->j|i} + \sum_{l\neq(i,j)} T_t^{l->j|i} > \sum_{l\neq(i,j)} T_t^{j->l|i} + R_t^{j->i} + D_t^{i,j}$ considering $\epsilon_t^{i,j}$ negligible.

Emigration flows assumption Here, the literature assume that, for all i, j such that $(\Delta S_{t+1})_{i\neq j} > 0$, then $\sum_{l\neq(i,j)} T_t^{l->j|i}$ i.e. transit flows are negligible with respect to $E_t^{i->j|i}$. In other words, they assume that what does the difference between the two sides of the equation is really emigration flows. Therefore, what happens is that positive differences on off-diagonal values - literally net inflows - are explained by movements from the diagonal to the column in question between t and t+1. Here, we consider that difference in total populations is negligible such that $\epsilon_t^{i,j}$ is very small and we do not need to care about birth data here since it would be relevant for looking at differences between the diagonals due to the nature of our matrices (bilateral migrant stock tables with place of birth in rows and place of residence in columns).

Alternative using diagonal values for arbitrage between transit/emigration flows We propose to condition the assumption of emigration flows on the observation of negative values on the diagonal (1); Since equal row margins, if we do not observe a negative delta on the diagonal, it must be observed off the diagonal and we would then assume transit flows instead of emigration flows (2); Incomplete cases.

$$\mathbf{\Delta S_{t+1}^{adj}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} -5.795 & 4.016 & 1.835 \\ -0.625 & 0.224 & 0.382 \\ 1.64 & 2.69 & -5.01 \end{pmatrix} \end{array}$$

Then we would assume that for the rows A and C that positive differences off the diagonal correspond to emigration flows. We would, nevertheless, assume that the positive value is due to transit flows for the row B.
Now that we proposed a methodology to deal with positive differences, we propose to follow with the interpretation of negative differences.

6.1.5 Negative differences: Net outflows

This is what we have in the case of negative differences on off-diagonal values:

$E_t^{i->j|i} + \sum_{l\neq(i,j)} T_t^{l->j|i} < \sum_{l\neq(i,j)} T_t^{j->l|i} + R_t^{j->i} + D_t^{i,j}$ considering $\epsilon_t^{i,j}$ negligible.

Return flows assumption The assumption made by the literature, in overall is that $\sum_{l\neq(i,j)} T_t^{j->l|i} + D_t^{i,j} + \epsilon_t^{i,j}$ is negligible compared to $R_t^{j->i}$. In other words, return flows dominate transit flows and death accounts.

Alternative using diagonal values for arbitrage between transit/return flows We propose to condition the assumption of return flows on the observation of positive values on the diagonal (1); Since equal row margins, if we do not observe a positive delta on the diagonal, it must be observed off the diagonal (since we start with trying to interpret a negative difference) and we would then assume transit flows instead of return flows (2); Incomplete cases.

$$\mathbf{\Delta S^{adj}_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} -5.795 & 4.016 & 1.835 \\ -0.625 & 0.224 & 0.382 \\ 1.64 & 2.69 & -5.01 \end{pmatrix} \end{array}$$

Then looking at row B, we would assume return flows combined with transit flows since we observe positive values on the diagonal and on other entries and since the difference at BA is the only negative difference.

6.1.6 To put it into a nutshell

The perfect setting for difference methods would be a couple of bilateral migrant stock tables with equal row margins and previously adjusted for demographic dynamics.
Let us be precise that having accounted for demographic dynamics removes one source of bias when trying to estimate flows out of differences in migrant stocks since death events could be counted as return or transit flows in the case of negative differences. Concerning birth events, it is not that it creates a direct source of bias as for death events since it increases values on the diagonals and we are not looking at them (dropping them whatever the approach used in difference-methods). But one thing we could think of is that it enables us to get closer to equal row margins i.e. no change in population by place of birth between t and t+1. Achieving such a goal, adjusting also for birth events, assuming a closed system implying some scaling approach, leads to the possibility to interpret any difference between the two migrant stock tables' cells as movements in columns. We can set our attention to a particular row and see the net dynamics by looking at where are the positive and negative differences have to cancel out since both stock tables were adjusted to have equal row margins.
This should have the following implications:

- Deduction of net return flows: If a difference in diagonal values is strictly positive meaning net inflows then it gives information on the existence of net return flows;
- Deduction of net emigration flows: If a difference in diagonal values is strictly negative meaning net outflows then it gives information on the existence of net emigration flows;
- Deduction of net transit flows: In the case of negative values on off-diagonal values combined with negative values on the diagonal value then these net outflows would rather interpret as net transit flows to balance with (necessarily) observed other cells with strictly positive values (since equal row margins);

Therefore, it raises one question, why does the literature skip the adjustment of bilateral migrant stock tables for demographic dynamics and equal row margins? Isn't it due to the sparsity of the matrices?

Let us give some illustrations of this alternative method:

$$\mathbf{S^{adj}_{t+1}} - \mathbf{S^{adj}_{t}} := \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 15 & 5 & 5 \\ 7.5 & 10 & 7.5 \\ 2 & 12 & 5 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 10 & 7.5 & 7.5 \\ 5 & 15 & 5 \\ 4 & 5 & 10 \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 5 & -2.5 & -2.5 \\ 2.5 & -5 & 2.5 \\ -2 & 7 & -5 \end{pmatrix} \end{array}$$

Then, keeping values on the diagonal to help our interpretation of these net positive and negative differences, we would consider that:
Considering the fact that total population by place of birth did not change over time (or at least we adjusted for that), we would observe that, with row A, that the native-born population increased in A and the foreign-A-born population decreased in B and C. Then, at least with the net flows we can detect from this difference and considering our system closed, one can reasonably interpret at least one negative cell as a source of net return flows. It could be possible indeed that 2.5 transited from C to B and then 5 returned from B to A. In fact we cannot say for sure and we still make assumptions. Let us say that the underlying assumption is less strong in particular cases that we are able to detect via diagonal values. Therefore this method could be improved maybe using additional information on the distribution at time t of the foreign born populations and applying some proportion method.

It would yield:

$$\mathbf{F_t^A} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 2 & 0 \\ 5 & 0 & 2.5 \\ 2.5 & 5 & 0 \end{pmatrix} \end{array}; \mathbf{F_t^{DN}} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 0 & 0 \\ 2.5 & 0 & 2.5 \\ 0 & 7 & 0 \end{pmatrix} \end{array}; \mathbf{F_t^{RN}} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 0 & 2 \\ 5 & 0 & 2.5 \\ 2.5 & 7 & 0 \end{pmatrix} \end{array}$$

where F_t^A contains estimates of transit flows while F_t^{DN} and F_t^{RN} does not distinguish between cases for values on the diagonal and simultaneous cases of negative values off the diagonal and on the diagonal along the same row.

Pros and Cons

Difference Method	Pros	Cons
"Steps: Adjust, Differentiate, Assign signs to type of flows based on diagonal values, Construct the matrix of flows out of it, Remove diagonal"	Partial Emigration, Transit and Return flows Estimation	Under-estimation due to cross flows conditional on place of birth (common limit to other methods)

6.2 Migration rates method

The intention of the migration rate approach as underlined in (Abel et al., 2019) is to estimate migration rates based on the foreign-born population's distribution across off-diagonal values of a given migrant stock table defined by place of birth criterion in the first part. The second part is to estimate the total flows between t and t+1 that occurred inside the system assuming a closed system. Finally, the goal is to obtain an origin and destination matrix concerned by the interval of time (t,t+1). We will see more precisely that it estimates an emigration flow matrix between t and t+1 assuming that total flows were the result of emigration flows only between t and t+1.

How does the literature address this? Do they proceed to adjustments of the bilateral migrant stock tables? How do they estimate the total flows between t and t+1?

In Abel's paper of 2019, in theory, migrant stock tables are adjusted for birth and death and there are equal row margins. In theory, an estimation of total flows would be then to look at the total of the differences between each column's margin between our couple of migrant stock tables to see how population varied between t and t+1 by each place of residence since, having adjusted for demographics and having total population stable in the system and assuming closed system, differences in columns must be explained by a mixture of emigration, transit, and return flows i.e. flows or movements in columns.
In practice, Abel did not implement on Migest (package on R for the estimation of the methods) an estimation of total flows based on a couple of migrant stock tables. It must probably be due to the sparsity of migrant stock tables and therefore he might have preferred to use data on net flows by country afforded by the World Population Prospects from the United Nations as a better source of information for estimating total flows.

Now the question is: what is really computed as migration rates in theory with a migrant stock table defined by place of birth criterion?

Notice that, ideally, we would have:

For i $\neq$ j, $\forall$ (i,j) $\in S := (A, B, C)$, $r_{i->j} := \frac{\sum_{k \in S} f_{i->j|k}}{\sum_{(i,j,k) \in (S^2 \setminus \Delta(SxS))xS} f_{i->j|k}}$

But, in practice, given a migrant stock table with the place of birth criterion, what is computed is the following:
$r_{i->j} := \frac{\sum_{k \in S} S_t^{i,k}}{\sum_{i,j|i \neq j} S_t^{i,j}}$

Now, to consider this as migration rates, there are multiple assumptions and remarks that can be underlined:

6.2.1 Assumption on negligible transit flows and stocks as past recent flows

- H1: The off-diagonal values are previous flows from t-1 to t in order then to consider these values as relevant information for determining the migration rate from i to j. Indeed, assume that half of the population on off-diagonal values in some places are old migrants according to the place of birth criterion - old in the sense they migrated between t-2 and t-1 for the most recent interval of time - then counting them in these places the migration rate would introduce a bias since we compute how all the flows between t and t+1 are distributed on off-diagonal values based on migrants that moved not the last period but on much earlier periods. Ideally, we would know also if they are recent migrants with respect to the length separating the censuses moments;

- H2: The transit inflows ($f_t^{k->j|i}$ with $k \neq (i,j)$ - where for k = i would account for emigration flow and k = j would account for total immobile foreign-population born in i living in j) is negligible in the total flows between t and t+1;

- Remark: We do not estimate out of this method any transit outflows or return flows (i.e. $f_t^{j->k|i}$ with $k \neq (i,j)$ - where for k = i would account for return flows and k = j would account for total immobile foreign-population born in i living in j). They are not at all estimated since migration rates are done based on $S_t^{i,j} > 0$ therefore emigration flows + transit inflows > transit outflows + return flows. Therefore migration rates are computed based on assumed emigration flows between t-1 and t strictly greater than transit outflows and return flows (while transit inflows are negligible) proxied by the foreign-born stock population at time t.

We remark H1 by considering for $i \neq j$ that $S_t^{i,j} = \sum_{k \neq j} f_{t-1}^{k->j|i}$ with a particular emphasize on the time index with:

$$r_{i->j} := \frac{\sum_{k \in S \setminus j} f_{t-1}^{k->j|i}}{\sum_{i,j|i \neq j} f_{t-1}^{i->j|i}}$$

We remark H2 by considering that, furthermore, with $i \neq j$ that $S_t^{i,j} = f_{t-1}^{i->j|i}$ i.e. we assume that $S_t^{i,j}$ is explained by direct emigration flows between t-1 and t, which gives:

$$r_{i->j} := \frac{f_{t-1}^{i->j|i}}{\sum_{i,j|i \neq j} f_{t-1}^{i->j|i}}$$

6.2.2 Presentation of the method with migrant stock tables defined by place of birth

To put into a nutshell what is computed, let us give a formula:

$$\hat{F}_t^{MR,(H1,H2)} := (S_t^{adj} - diag(S_t^{adj})) \times \frac{M}{s(S_t^{adj} - diag(S_t^{adj}))}$$

$$s(S_t^{adj} - diag(S_t^{adj})) := Tr((S_t^{adj} - diag(S_t^{adj}))\iota); \ \iota := \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Let us compute an example following the migrant stock table we adjusted in the previous section S_t^{adj}. We will also use S_{t+1}^{adj} for estimating total flows between t and t+1.

$$\mathbf{S_t^{adj}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 \\ 4.5 & 12.2 & 4.375 \\ 3.84 & 1.536 & 12.6 \end{pmatrix} \end{array} \Rightarrow \mathbf{\hat{F}_t^{MR}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & \frac{3.2 \times \hat{M}}{20.951} & \frac{3.5 \times \hat{M}}{20.951} \\ \frac{4.5 \times \hat{M}}{20.951} & 0 & \frac{4.375 \times \hat{M}}{20.951} \\ \frac{3.84 \times \hat{M}}{20.951} & \frac{1.536 \times \hat{M}}{20.951} & 0 \end{pmatrix} \end{array}$$

Then, estimating M by proceeding likewise:

$$\mathbf{S_{t+1}^{adj}} - \mathbf{S_t^{adj}} := \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 5.775 & 6.6 & 4.95 \\ 3.828 & 12.441 & 4.785 \\ 6 & 4 & 8 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 \\ 4.5 & 12.2 & 4.375 \\ 3.84 & 1.536 & 12.6 \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \\ Total \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} -5.725 & 3.9 & 1.45 \\ -0.672 & 0.241 & 0.41 \\ 2.16 & 2.464 & -4.6 \\ -4.37 & 6.605 & -2.74 \end{pmatrix} \end{array}$$

In Abel's paper, the estimated **total net flows** is computed then by adding the sum of the absolute differences in

total population by place of residence. There are assumptions behind this choice. In practice, we take world population prospects' data for estimating total flows. Nevertheless, let us illustrate this method: $\hat{M}$ = (approximately) $2 \times 6.605 = 13.21$

The resulting estimated flow table is:

$$\hat{\mathbf{F}}_{\mathbf{t}}^{\mathbf{MR}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & \frac{3.2\times13.21}{20.951} & \frac{3.5\times13.21}{20.951} \\ \frac{4.5\times13.21}{20.951} & 0 & \frac{4.375\times13.21}{20.951} \\ \frac{3.84\times13.21}{20.951} & \frac{1.536\times13.21}{20.951} & 0 \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 2.016 & 2.205 \\ 2.835 & 0 & 2.756 \\ 2.41 & 0.96 & 0 \end{pmatrix} \end{array}$$

Remark In that case, zero-estimation of total flows may be due to periods were emigration flows are exclusive and equal.
Take this example:

$$\mathbf{S}_{\mathbf{t+1}}^{\mathbf{adj}} - \mathbf{S}_{\mathbf{t}}^{\mathbf{adj}} := \begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} 15 & 0 \\ 0 & 15 \end{pmatrix} \end{array} - \begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} 10 & 5 \\ 5 & 10 \end{pmatrix} \end{array} = \begin{array}{c} \\ A \\ B \\ Total \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} 5 & -5 \\ -5 & 5 \\ 0 & 0 \end{pmatrix} \end{array}$$

This is an extreme case. Nevertheless, clearly in that case, with adjusted bilateral migrant stock tables, one could simply take absolute values of differences off the diagonal to estimate total flows which would give 10. Of course this is always an under-estimation due to cross flows conditional on place of birth bias effect.

Back to the previous example, we could add 5.725, 0.672 and 4.6 to be pretty close to 13.21. It would be the sum of net flows for each rows with of course many underlying assumption and an under-estimation of all the flows our migrant stock tables cannot detect.

6.3 Contingency Table Methods

6.3.1 Motivation

One of the main advantages of this procedure is that it enables us to estimate emigration, return, and transit flows based on migrant stock data.

6.3.2 Poisson Regression setting with SIM model

In (Abel, 2013), the cells are estimated according to a Poisson regression setting. We present the general specification with an independence assumption between the net flows (always since due to cross flows birth-place specific bias effect we cannot in particular estimate total flows by matching with migrant stock data defined by place of birth criterion not by previous place of residence) and with quasi-independence assumption with constraints on the diagonal.

Model specification 1: "Independence"
Assume $F_{ij}^k \sim \text{Poisson}(y_{ij}^k)$ with $y_{ij}^k = \alpha_i^k \ \beta_j^k$ (with offset term at 1).

Model specification 2: "Migrant minimization"
Assume $F_{ij}^k \sim \text{Poisson}(y_{ij}^k)$ with $y_{ij}^k = \alpha_i^k \ \beta_j^k \ \delta_{ijk}^{I(i=j)}$ (with offset term set at 1).

where parameters α_i^k controls for the place of birth (with index k)-place of origin (with index i), β_j^k controls for the place of birth (with index k)-place of destination (with index j) and $\delta_{ijk}^{I(i=j)}$ is set in model specification 2 to set the maximal values on the diagonal, which is what we will introduce in the next part of the report.

6.3.3 "Independence" Estimation

We start with:

$$\mathbf{S^{adj}_{t+1}} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 5.775 & 6.6 & 4.95 \\ 3.828 & 12.441 & 4.785 \\ 6 & 4 & 8 \end{pmatrix} \end{array}; \mathbf{S^{adj}_{t}} = \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 11.5 & 3.2 & 3.5 \\ 4.5 & 12.2 & 4.375 \\ 3.84 & 1.536 & 12.6 \end{pmatrix} \end{array}$$

You define your birth-place specific bilateral migrant flow tables with matching adjusted migrant stock data constraint.

$$\mathbf{F^A_t} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} \alpha^A_A\beta^A_A & \alpha^A_A\beta^A_B & \alpha^A_A\beta^A_C & S^{adj}_{t;A,A} \\ \alpha^A_B\beta^A_A & \alpha^A_B\beta^A_B & \alpha^A_B\beta^A_C & S^{adj}_{t;A,B} \\ \alpha^A_C\beta^A_A & \alpha^A_C\beta^A_B & \alpha^A_C\beta^A_C & S^{adj}_{t;A,C} \\ S^{adj}_{t+1;A,A} & S^{adj}_{t+1;A,B} & S^{adj}_{t+1;A,C} & \end{pmatrix} \end{array}; \mathbf{F^B_t} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} \alpha^B_A\beta^B_A & \alpha^B_A\beta^B_B & \alpha^B_A\beta^A_C & S^{adj}_{t;B,A} \\ \alpha^B_B\beta^B_A & \alpha^B_B\beta^B_B & \alpha^B_B\beta^A_C & S^{adj}_{t;B,B} \\ \alpha^B_C\beta^B_A & \alpha^B_C\beta^B_B & \alpha^B_C\beta^B_C & S^{adj}_{t;B,C} \\ S^{adj}_{t+1;B,A} & S^{adj}_{t+1;B,B} & S^{adj}_{t+1;B,C} & \end{pmatrix} \end{array}$$

$$\mathbf{F^C_t} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array}\begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} \alpha^C_A\beta^C_A & \alpha^C_A\beta^C_B & \alpha^C_A\beta^C_C & S^{adj}_{t;C,A} \\ \alpha^C_B\beta^C_A & \alpha^C_B\beta^C_B & \alpha^C_B\beta^C_C & S^{adj}_{t;C,B} \\ \alpha^C_C\beta^C_A & \alpha^C_C\beta^C_B & \alpha^C_C\beta^C_C & S^{adj}_{t;C,C} \\ S^{adj}_{t+1;C,A} & S^{adj}_{t+1;C,B} & S^{adj}_{t+1;C,C} & \end{pmatrix} \end{array}$$

Then for k in (A,B,C), we proceed to the estimation of the parameters $(\alpha^k_i, \beta^k_j)_{(i,j)\in(A,B,C)}$ with iterative proportion fitting procedure (see annex part). The values of convergence correspond to the MLE estimates. We run the procedure on R via the package Migest to obtain the estimates and we introduce the results just below. We used the diff demo method implemented by Abel and used in his works of 2019.

$$\hat{\mathbf{F}}^{\mathbf{A}}_{\mathbf{t}} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 3.75 & 4.1 & 3.1 \\ 1.1 & 1.2 & 0.91 \\ 1.18 & 1.29 & 0.97 \end{pmatrix} \end{array}; \hat{\mathbf{F}}^{\mathbf{B}}_{\mathbf{t}} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0.8 & 2.6 & 1 \\ 2.3 & 7.59 & 2.9 \\ 0.77 & 2.53 & 0.96 \end{pmatrix} \end{array}$$

$$\hat{\mathbf{F}}^{\mathbf{C}}_{\mathbf{t}} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 1.32 & 0.88 & 1.77 \\ 0.51 & 0.34 & 0.69 \\ 4.03 & 2.70 & 5.40 \end{pmatrix} \end{array}$$

Summing over the place of birth and removing diagonal values since stayers, we obtain an origin and destination matrix estimation:

$$\hat{\mathbf{F}}_{\mathbf{t}} := \begin{array}{c} \\ A \\ B \\ C \end{array}\begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 8.14 & 6.14 \\ 3.65 & 0 & 4.26 \\ 5.9 & 6.63 & 0 \end{pmatrix} \end{array}$$

Abel published his estimates on the website Figshare where the code and the data are available. All the procedure and the code is free to access.
One good trick when the offset term is set to 1 in order to obtain the values of the matrix is to do: $F^k_{t;i,j} = \frac{S^{adj}_{t;k,i} \times S^{adj}_{t+1;k,j}}{\sum_l S^{adj}_{t;k,l}}$. In other words, multiply the total population of individuals born in k living at time t in i by the total population of individuals born in k living at t+1 in j and divide this quantity by the total population of the system that we obtained after adjustments.

6.3.4 "Migrant Minimization" Estimation

General setting:
Without loss of generality, fix k = A.

$$\mathbf{F_t^A} := \begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} min(S^{adj}_{t;A,A}, S^{adj}_{t+1;A,A}) & \alpha^A_A\beta^A_B & \alpha^A_A\beta^A_C & S^{adj}_{t;A,A} \\ \alpha^A_B\beta^A_A & min(S^{adj}_{t;A,B}, S^{adj}_{t+1;A,B}) & \alpha^A_B\beta^A_C & S^{adj}_{t;A,B} \\ \alpha^A_C\beta^A_A & \alpha^A_C\beta^A_B & min(S^{adj}_{t;A,C}, S^{adj}_{t+1;A,C}) & S^{adj}_{t;A,C} \\ S^{adj}_{t+1;A,A} & S^{adj}_{t+1;A,B} & S^{adj}_{t+1;A,C} & \end{pmatrix} \end{array}$$

In the case of minimization, we add an assumption on the stayers (i.e. the values on the diagonal of the bilateral flow migrant table), as follows:
$F^k_{t;i,i} = \min(S^{adj}_{t;k,i}, S^{adj}_{t+1;k,i})$. This yields, applied to our example used all along the report:

$$\begin{array}{c} \\ A \\ B \\ C \\ \sum_j \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & \sum_i \end{array} \\ \begin{pmatrix} 5.775 & \alpha^A_A\beta^A_B & \alpha^A_A\beta^A_C & 11.5 \\ \alpha^A_B\beta^A_A & 3.2 & \alpha^A_B\beta^A_C & 3.2 \\ \alpha^A_C\beta^A_A & \alpha^A_C\beta^A_B & 3.5 & 3.5 \\ 5.775 & 6.6 & 4.95 & \end{pmatrix} \end{array}$$

We just put zeros where they must be to respect margin constraints and set values to match margins and finally set at zeros the diagonal values to extract origin/destination estimates. We let R do it for us with diff demo through Migest package.

$$\hat{\mathbf{F}}_\mathbf{t}^\mathbf{A} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 3.6 & 1.55 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{array}$$

You can see the minimization effect of adding maximal values on the diagonal. As underlined by (Azose and Raftery, 2019) this is due to the underlying assumption that moving is very costly and people tend to stay at their current place. In their paper, they talk of a "lower bound" estimate.
We also afford the resulting origin and destination matrix thanks to Migest package that we obtain by summing over the birth-place specific estimated bilateral migrant flow tables from which we removed diagonal values to focus on estimated flows:

$$\hat{\mathbf{F}}_\mathbf{t} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} 0 & 4.314 & 1.74 \\ 0 & 0 & 0 \\ 2.19 & 2.548 & 0 \end{pmatrix} \end{array}$$

6.3.5 Mixed Estimation

Convex combination of Migrant Minimization (MM) and Independence (I) estimates with parameter w. But which w to choose? We present the methodology used by (Azose and Raftery, 2019).

Choice of a dataset of "true flows" They choose IMEM and OECD data covering a period of their flow estimates. They extract four types of reference data from it, respectively Inflows to OECD countries divided by the population of the destination country, Outflows to OECD countries divided by the population of the origin country, IMEM flows divided by the population of the origin country/divided by population of the destination country.
Their estimates are based on migrant stock data based on the definition by place of birth while OECD data contains nationality-based defined migrants.
The data of the OECD and IMEM are yearly while their estimates are quinquennial. See strong assumptions to consider how they proceed to comparisons and error computations.

Choice of a loss function They choose a quadratic loss function with squared scaled errors. The scale parameter is to obtain a rate-scale by diving flow by the population of origin. It is done to avoid the downsides of squared log differences or squared differences such as huge errors for small differences or identical errors for the same order of difference without taking into account the order of magnitudes. $L(f_{ij}, \tilde{f}_{ij}) := (\frac{f_{ij} - \tilde{f}_{ij}}{population_i})^2$

Strong assumptions on Populations and Migrations

- Flows within each five-year time period occur uniformly over time;
- The five-year transition counts are equal to the sum of the one-year transition counts;
- Births and deaths were correctly accounted for so that all population change is due to migration;
- Neither country populations nor the composition of migrant stocks will change too dramatically over the long time period.

Solving Easy Convex Optimization Setting $\underset{w \in [0,1]}{argmin} ||M_t - (w\tilde{M}_t^I + (1-w)(\frac{1}{5}\tilde{M}_t^{MM})||_F^2$

"They obtained w = 0.87 with European migration flows between 2002 and 2008 from the IMEM project." (Abel et al., 2019). This was re-used by Abel in his paper of 2019.

7 Sources of bias

Commenting on recent advances in research on the estimation of migration flows, Willekens asserted "These are not flow data, but they come close." (Willekens et al., 2016)

7.1 No adjustment as a source of bias in the case of complete migrant stock tables

No adjustment could be a source of limit for each of the methods as far as certain associated assumptions are concerned.
We will show it in the next section more precisely, but let us list what could happen for each method.

In general: After demographically adjusting the migrant stock tables, the goal of obtaining equal row margins is to interpret differences in columns as net flows targeted on each sub-population by place of birth criterion.

Difference method: Assuming off-diagonal net outflows (negative differences) are return flows while the net outflows might be explained by the fact that death + return + transit > emigration flows. We might want to adjust for death data to estimate (return + transit - emigration) by doing the difference between the two stock tables and then assuming return flows are dominant among return + transit.

Migration rate method: If it depends on how to estimate migration rates, it does not seem to be necessary to adjust with death or birth data. Nevertheless, if we try to estimate the total flows with the couple (S_t, S_{t+1}) by taking the sum of the absolute differences in total population by place of residence as suggested by (Abel et al., 2019), the assumption is that differences are explained by flows, therefore one needs at least to verify that the two stock tables have the same total population in it and for it to adjust for demographic dynamics. Then, the goal is to obtain equal row margins to interpret differences in columns as net flows by targeted population with the place of birth (fixing row/place of birth) and therefore column margins' (fixing column/place of residence) differences as net flows by summing over all targeted flows (net flows for foreign-pop from A, net flows foreign-pop from B for off-diagonal values and on the diagonal net flows for return + transit - emigration since birth removed after demographic adjustment).

7.2 Classical measurement bias

It relates to the methodologies of data collection and dissemination by national statistical institutions then gathered and rearranged by international institutions such as the World Bank or the United Nations.

7.3 Sparse migrant stock tables in practice

Migrant stock tables data are available with intervals of time of ten years from the World Bank databases and with intervals of time of five years from the United Nations. The matrices are very sparse. For example, the results of Abel in his 2019 paper are based on migrant stock tables from the United Nations, counting all the migrant stock tables, where 76 % of entries with values set to zero. This is, of course, a huge source of bias to estimate true flows.

To account for it, the literature mentions notably "low wage differentials, large distances, high migration or visa costs." (Beine et al., 2011). Nevertheless, due to the scope of it, it must be due to a lack of data/measurements.

7.4 Undetected cyclical and seasonal migrant flows

From these estimation methods, we omit to count cyclical and seasonal flows inside the time interval of concern. Consider a couple of migrant stock tables S_t, S_{t+h} with a length of time h separating them. Therefore, any agent that simultaneously emigrated and returned back, strictly between t and t+h, is undetected. It means in particular that if the goal is to estimate short-term flows, the bias is obvious. Short-term flows can include cyclical and seasonal migrant flows. By cyclical flows, we could include any weekly, monthly, or any flow-t- loop repeated as many times as possible in this strict interval of time due to working activities for example. By seasonal flow, we could think of any migrant flows economically related to season-dependent economic activities in a specific subset of countries of our system.

7.5 Cross flows by place of birth bias effect

This bias is simply to consider cross flows conditional on the place of birth occurring in the corresponding time interval between any couple of countries of our system.

If between t and t+h, the total flow $f_{i->j|k}$ from i to j for people born in k is balanced with $f_{j->i|k}$ from j to i for people born in k, then, ceteris paribus, there is no change in $S_{t,k,i}$ and $S_{t+h,k,j}$.
In fact, the only difference we can detect is as underlined before net flows conditional on the place of birth between any pair of countries in our system.

7.6 Restriction to the place of birth criterion

Even though the place of birth criterion is mostly used due to the resulting property of comparability for off-diagonal values, we miss migrants by citizenship or recent change of place of residence.

8 Potential sources of research

8.1 Dealing with structural limits

8.1.1 Sparsity of matrices

The fact that the matrices are very sparse implies that adjusting for demographic dynamics like with death data by place of residence is not a good approach since, in the worst case, it can produce negative entries to adjusted bilateral migrant stock tables. It can create artificial net emigration flows in this extreme case which is not the initial goal of the adjustment. Notice that it is particularly true if at t+1 off-diagonal cell's value is at zero. Notice also that zeros at time t, using uniform imputation method, then create negative entries at time t and ensure artificial net emigration flows. Nevertheless, using the proportion imputation method, we still have to deal with zero entries at time t+1 and the impact it can have on our estimates. For example, in that case, it would reduce the amount of net return flows.
The impact of this fact is still to be studied on how we can improve difference methods or migration rate methods by adjusting these sparse migrant stock tables.
The goal is indeed to relax the strong assumption done when interpreting negative differences as net return flows. On the contrary, with a complete matrix and adjustment for death data while assuming equal row margins and a closed system, one can interpret negative differences as transit or return flows depending then on the difference value on the diagonal. For example, in that scenario, take the case of a positive difference value on the diagonal which then implies net return flows. Therefore, since equal row margins, you expect that one other place of residence is negative for the delta. That is where you will deduce that it produced net return flows. On the contrary, if we had a negative difference value on the diagonal, it would be less obvious to assert that another negative delta on the same row corresponds to a net return flows - but would rather be interpreted as net transit outflows.

8.1.2 Advantage of completing matrices for adjustment strategies

There are some possible strategies to implement once the sparsity of matrices is solved:

- Difference methods on demographically adjusted migrant stock tables combined with migration rates to obtain estimates of emigration, transit, and return flows;
- Difference methods on completely adjusted migrant stock tables and interpretation of positive and negative differences depending on the value on the diagonal of the difference matrix and via a migration rate approach;
- Migration rates mechanically improved with adjusted migrant stock tables and potentially estimating total flows via a couple of adjusted migrant stock tables;

8.1.3 Deepen sensitivity analysis on the different methods available

Open for reflection. This need is underlined by the literature: see (Berlemann and Steinhardt, 2017).

8.2 Method selection strategy

8.2.1 Selecting according to targeted migrant population

Open for reflection. Thinking about refugee data.

8.2.2 Selecting via correlation matrices with existing raw flow data

Open for reflection. See (Abel et al., 2019) and the report on climate migration of Peter Kamal supervised by Anouch Missirian 2022, where they used UNHCR flow data to compare with their estimates obtained by difference-methods.

9 Glossary

Migration:

We focus on the migration of human populations. Nevertheless, this phenomenon concern all levels of biological organizations. The biological meanings of migration are not necessarily of interest in this project since they

would require, to be identified during the data collection and the estimation of the variables, less-than-a-year-data, micro-data, and a finer geographical grid ((Dingle and Drake, 2007)). The adopted approach is quantitative, macro, global, and bilateral. Migration is yet to be defined rigorously in order to obtain targeted measurements of the macro phenomenon. In the presentation of the following estimation methods, migration will be considered following the place of birth criterion i.e. people identified as migrants in a given country due to a difference in place of birth. We do not focus on human migration at a micro-level ((Hägerstrand, 1969)) but rather on migration as a macro-phenomenon, an aggregated phenomenon as the sum of all the migrations at an individual level. In the best case, we would like to disaggregate this macro-phenomenon by age and sex. The approach of Hagerstrand would have led to a focus on migrations at the finest geographical and individual levels. Discrete trajectories give the main picture in a 3d space with two spatial dimensions and a time dimension. Horizontal movements correspond to migrations or changes of house while vertical movements correspond to the length of time staying at the corresponding place before moving again;

Sparse matrix:
Matrix with entries mostly equal to zero.

10 Annex

10.1 Poisson Regression with expectation defined with SIM

10.1.1 Existence of MLE in Poisson Regression

The problem of the existence of the maximum likelihood estimator of Poisson regression is addressed by the literature and we refer to the following article for curious readers: (Silva and Tenreyro, 2010).
They notably emphasize that the MLE does not exist in the case of multicollinearity of the regressors. In the case of the model specifications proposed by Abel with a simple interaction model, the only regressors are constant parameters accounting for the place of birth-specific effect, origin-specific effect, and destination-specific effect. Therefore if the model is well specified under this basic framework, the existence of MLE should be guaranteed.
For more complex model specifications, if multicollinearity there is, the literature affords solutions if so: (Lukman et al., 2021).

10.1.2 Margin data: sufficient statistics for MLE estimators

Consider F_{ij}^k the entries of a birth place-specific bilateral migrant flow table with fixed margin constraints.
Assume $F_{ij}^k \sim \text{Poisson}(y_{ij}^k)$ with $y_{ij}^k = \alpha_i^k \ \beta_j^k \ \delta_{ijk}^{I(i=j)}$. We set the offset term to 1 without loss of generality.
Solve:

$$\underset{(\alpha_i^k,\beta_j^k,\delta_{i,j,k})_{(i,j)}}{\text{argmax}} \quad log(P(\bigcap_{(i,j)}(F_{ij}^k = f_{ij}^k)))$$

$$\Leftrightarrow \underset{(\alpha_i^k,\beta_j^k,\delta_{i,j,k})_{(i,j)}}{\text{argmax}} \quad log(\prod_{(i,j)} \frac{e^{-y_{ij}^k}(y_{ij}^k)^{f_{ij}^k}}{(f_{ij}^k)!}) \text{ by independence;}$$

$$\Leftrightarrow \underset{(\alpha_i^k,\beta_j^k,\delta_{i,j,k})_{(i,j)}}{\text{argmax}} \quad \sum_{(i,j)}(f_{ij}^k log(\alpha_i^k\beta_j^k\delta_{ijk}^{I(i=j)}) - log(f_{ij}^k!) - \alpha_i^k\beta_j^k\delta_{ijk}^{I(i=j)}).$$

We define $\text{F} := \sum_{(i,j)}(f_{ij}^k log(\alpha_i^k\beta_j^k\delta_{ijk}^{I(i=j)}) - log(f_{ij}^k!) - \alpha_i^k\beta_j^k\delta_{ijk}^{I(i=j)})$.
We take the first derivatives with respect to each parameter. We show that row and column margins are sufficient statistics.

$\forall i \frac{\partial F}{\partial \alpha_i^k} = 0$
$\Leftrightarrow$
$\forall i, \frac{\sum_j f_{ij}^k}{\hat{\alpha}_i^k} - \sum_j \hat{\beta}_j^k \hat{\delta}_{ijk}^{I(i=j)} = 0 \Leftrightarrow$
$\hat{\alpha}_i^k = \frac{f_{i.}^k}{\sum_j \hat{\beta}_j^k \hat{\delta}_{ijk}^{I(i=j)}}$
$\forall j \frac{\partial F}{\partial \beta_j^k} = 0$
$\Leftrightarrow$
$\forall j, \frac{\sum_i f_{ij}^k}{\hat{\beta}_j^k} - \sum_i \hat{\alpha}_i^k \hat{\delta}_{ijk}^{I(i=j)} = 0 \Leftrightarrow$
$\hat{\beta}_j^k = \frac{f_{.j}^k}{\sum_i \hat{\alpha}_i^k \hat{\delta}_{ijk}^{I(i=j)}}$
$\forall i \frac{\partial F}{\partial \delta_{iik}} = 0$
$\Leftrightarrow$
$\forall i, \frac{f_{ii}^k}{\hat{\delta}_{iik}} - \hat{\alpha}_i^k \hat{\beta}_i^k = 0 \Leftrightarrow$
$\hat{\delta}_{iik} = \frac{f_{ii}^k}{\hat{\alpha}_i^k \hat{\beta}_i^k}$.

10.1.3 Iterative proportional fitting procedure

Take a contingency table with fixed consistent margin constraints (i.e. sum over the row margins equal sum over column margins).

In the case of the specification without constraints on the diagonal and no offset term:

$F_{ij}^k \sim \text{Poisson}(y_{ij}^k)$ with $y_{ij}^k = \alpha_i^k \ \beta_j^k$.

Then, estimating by MLE, we obtain:

$\forall i, \hat{\alpha}_i^k = \frac{f_{i.}^k}{\sum_j \hat{\beta}_j^k}$
$\forall j, \hat{\beta}_j^k = \frac{f_{.j}^k}{\sum_i \hat{\alpha}_i^k}$

Then, the iterative proportional fitting procedure is simply to fix some initial values at $(\beta_j^0)_{j\in(1,..,n)}$ such that we can compute associated values for $(\alpha_i^0)_{i\in(1,..,n)}$. Then we continue with $(\beta_j^1)_{j\in(1,..,n)}$ and so forth... such that we obtain a recursive sequence of order 1 of values for the couples $(\alpha_i^t \beta_j^t)$ where t correspond to the iteration step.

To visualize this, one takes a look at the contingency table and thinks of the estimated values obtained by MLE. Then assign initial values on the β's or the α's and then sum over the rows (respectively the columns) to match with margin constraints. The next step is to set back as unknowns the β's or the α's and to sum over the columns (respectively the rows) to match margin constraints. At that moment stop and compare with previous values of the parameter. If very small difference you can stop, otherwise, continue!
Do we have proof that it necessarily converges? Well, there exist papers about that, see for example: (Ruschendorf, 1995).

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